

CARGOMATE
SMART ROBOTIC INVENTORY SYSTEM WITH CLOUD
INTEGRATION

25J-26J-032

Project Proposal Report

D K D G Perera

BSc. (Hons) Degree in Information Technology Specialized in Computer
Systems and Network Engineering

Department of Computer Systems Engineering

Sri Lanka Institute of Information Technology

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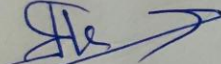
August 2025

Declaration of the Candidate & Supervisor

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any previously published or written by another person material, except where the acknowledgement is made in the text.

Group Member Name	Student ID	Signature
Perera D K D G	IT22312594	

The above candidate are carrying out research for the undergraduate Dissertation under my supervision.


Signature of the Supervisor :

Date: 30/08/25

Abstract

In order to improve the accuracy, efficiency and automation of warehousing processes, this project recommends the development and deployment of CargoMate, a Smart Robotic Inventory System with Cloud Integration. CargoMate is a cost-effective yet scalable system for modern inventory management that combines smart handling of goods, autonomous robotic mobility, and real-time cloud monitoring.

CargoMate is able to navigate around the warehouse with accuracy and autonomously since it is equipped with a navigation system. CargoMate has accurate movement along planned routes through the combination of RFID (Radio Frequency Identification) junction detection, Ultrasonic obstacle detection, and infrared sensor line track following. If there is a path loss, there is a recovery process to enable it to detect and realign itself, providing consistent and reliable performance even in dynamic environments.

CargoMate includes automatic inventory management, including selecting, ordering, and moving goods to points of destination, and navigation. Centralized supervision, task administration, and operational information through Firebase cloud integration assist warehouse managers by enabling synchronized navigation and inventory information in real time.

Path accuracy, junction detection, collision avoidance, recovery effectiveness, and inventory management were validated for CargoMate in simulated warehouse environment. CargoMate illustrates how effective inventory management and precise navigation can be combined with inexpensive, high-quality, and large-scale robotic technology to improve warehouse operation automation and efficiency.

Key words: CargoMate, RFID junction detection, Smart Robotic Inventory System, Ultrasonic obstacle detection, recovery process

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1.INTRODUCTION

1.1 Background Study

For warehouse robots operating autonomously, navigation quality is crucial since it provides useful mobility within sealed spaces with minimal human input. Conventional industrial robot navigation systems utilize advanced sensors like LiDAR (Light Detection and Ranging) or computer vision systems, which are very precise but profoundly expensive and computationally demanding. The demand for hardy, low-cost, and lightweight solutions that can promise efficient navigation without costly infrastructure for small- and medium-sized warehouse automation is escalating.

Infrared (IR) line-tracking is a standard method in which robots track a light-colored tape line across the floor. The method is suitable for repetitive warehouse use due to it being simple, cheaply priced, and reliable in controlled indoor settings. Yet, one drawback of traditional line-followers is that they depend on an unbroken path visibility and do not have backup plans in case the line is lost.

Radio Frequency Identification (RFID) has been investigated as an adjunct navigation aid to address decision-making challenges at intersections. RFID tags installed at crucial intersections offer robots location intelligence, allowing them to differentiate between alternatives and execute task-specific behavior. RFID is reliable under differing light conditions and does not require heavy computation capacity, unlike vision-based or SLAM-based junction detection.

Obstacle detection and evasion are a critical aspect of travel within warehouses.

Ultrasonic sensors have been most frequently utilized for near-range object detection because of their affordability and ease of integration. Although they give consistent close-range detection, the incorporation in addition to line following ensures the robot has a simple time returning to its intended path and avoid running into unforeseen obstructions.

Lastly, recovery mechanisms need to be brought in for improving robustness. Mechanisms of self-correction enable the robot to halt, inspect its surroundings, and re-synchronize with the navigation path without interference in scenarios where it gets lost due to sensor noise, surface undulations, or environmental interference. Continuity of operation and autonomy are tremendously provided by this capability.

Modern robotic systems are able to provide robust and economical navigation appropriate for warehouse environments employing these components: ultrasonic obstacle detection for protection, RFID-based intersection detection for decision-making, recovery methods for fault-tolerance, and infrared line-tracking for structured navigation. CargoMate's navigation system design is based on such a combined approach.

1.2 Literature Survey

Path following, junction/waypoint localization, and local safety maneuvers (obstacle avoidance/detection) are generally combined in warehouse robot navigation. Low-cost solutions generally use RFID indoor location semantics, IR reflectance sensors for line tracking, and ultrasonic ranging for nearby threats. Lightweight IoT/cloud services are occasionally added for task allocation and supervision. Publications [1]–[6] collectively cover each of these elements.

IR Line Following (Path Tracking)

They demonstrate reflectance-based tracking to be reliable on structured tracks and appropriate for low-cost platforms by applying and designing a line-follower with obstacle detection using low-cost sensors and embedded control, as stated in [1]. In contrast to tape-based guidance, they take note of sensor calibration, thresholding, and robust motor actuation for curves and turns. This work is extremely relevant and applicable to CargoMate since determination of IR line following will be successful depending on it.

RFID-Based Junction Detection

When GPS is not available, RFID has been suggested over the last few decades for indoor location and direction finding. A concept of navigation based on RFID for indoor mobile robots is illustrated in [4], emphasizing infrastructure simplicity and read tag reliability. A guided prototype using RFID placement is seen in [2], and environment-mounted tags can encode context and location at low onboard processing. They investigate RFID signal intensity sensing as a localization signal in [3] and demonstrate that RF signals can be used for navigation decisions without vision/LiDAR stacks. Firebase is used in CargoMate to map each tag's ID with task logic (pick, put, and go on) after the placement of RFID tags at intersections.

Ultrasonic Obstacle Detection & Avoidance

They confirm threshold-triggered halting and basic detours as viable for near-field threats in [5] using ultrasonic-based sensing and evading on a mobile robot. Expanding this to an IoT-focused design in [6], employing Arduino + ultrasonic, they show how connectivity can enable monitoring and telemetry advantage without requiring much onboard computer horsepower. By means of a four-step procedure—stop → temporary reversal → sidestep → IR-guided re-approach to the black line—CargoMate establishes a clear distance limit of approximately 20 cm.

1.3 Research Gap

Table 1.3: Table of the Research Gap

Features	Current Research	Identified Gap	Proposed Solution
IR Line Following (Path Tracking)	A line follower robot with basic obstacle detection and selection from a set of color-coded paths is	It is not equipped with cloud connectivity, RFID-based junction semantics, or heavy line-loss recovery	CargoMate has three infrared sensors that trace a black tape. Once the line is lost, it goes into recovery

	demonstrated by [7]. A closed-loop feedback method is employed for students to follow through narrow curves.	tailored for warehouse aisles but instead is concerned with the selection of color and general line following.	mode and spins around to get back in line.
RFID-Based Junction Detection	Mobile robot navigation in a warehouse setup is explored in [8], with improved A* path generation and extras like battery health monitoring introduced.	Lacks cloud integration and firm RFID tag-based junction detection for task logic at endpoints (i.e., pick/place actions).	At junctions, CargoMate reads RFID tags, matches IDs with Firebase, and takes agreed actions such as picking or releasing items.
Ultrasonic Obstacle Avoidance & Detection	They suggest in [9] applying partially sampled 3D point clouds from a curved LiDAR to predict and classify obstacles to support real-time AGV intelligence with common onboard processors.	Too luxurious for a low-budget warehouse robot, it applies costly LiDAR sensors and compute-heavy calculation. fails to investigate lightweight sidestep + rejoin-line approaches based on cheap sensors.	CargoMate senses obstacles in 20 cm with an ultrasonic sensor, then comes to a halt, reverses, goes around, and re-melines along the black line to successfully handle obstacles.

1.4 Research Problem

This is the research question investigated in this project.

How do warehouse robots end up with a low-cost, cloud-based navigation system integrating RFID-based junction detection, ultrasonic collision avoidance, and infrared-based line-following with recovery to provide reliable, scalable, and fault-tolerant inventory operations?

Depending on that the research question can be broken into some aspects:

1. Limitations of IR Line-Following

While traditional infrared line-following systems operate decently enough in following straight paths, they are fairly fragile in the practical warehouse setting. They become problematic with sudden turns, line breaks, and surface noise in particular. Furthermore, the majority of described implementations lack a systematic line recovery strategy, which makes them semi-autonomous at best in the case of lost path.

2. Poor Intelligent Junction Management

Modern navigation systems typically resort to computationally costly SLAM approaches or pre-defined rule sets for crossing decisions. While these are not adaptable to dynamic warehouse task reassignment, however. Robots are unable to dynamically reallocate their routes based on stock requirements since no reliable and lightweight junction detection method (e.g., RFID tags associated with a cloud infrastructure) exists. Cost vs.

3. Complexity of Obstacle Avoidance

Although extremely accurate obstacle recognition techniques such as LiDAR-based point cloud processing exist, their computation and expense make them impractical for small- and medium-scale warehouse operations. Low-cost ultrasonic sensors, however, are useful in stationary situations since they tend to be deployed individually without advanced stop-sidestep-rejoin methods.

4. Connectivity to Cloud Platforms

Most existing works emphasize autonomous robotic movement that is not real-time synchronized with cloud infrastructures. This causes issues for multi-robot coordination in extended warehouse environments, bounded scalability, and an absence of centralized monitoring.

5. Full reliability

Rather than approaching line-following, junction detection, and obstacle avoidance as a unified navigation system, most previous work tends to study these effects independently. Under real-world warehouse conditions, where every problem occurs simultaneously, this piecemeal methodology fails to accommodate smooth implementation.

In order to fix these problems, this project has tangible applications to warehousing processes as well as to academic research, with the idea of showing that low-cost robots can be redirected in order to solve actual logistic problems.

2. OBJECTIVES

2.1 Main Objective

To create a smarter warehousing automation solution that combines cloud-based inventory management, robotic put away and picking, autonomous navigation, and QR-code based product identification to augment operational efficiency, accuracy, and scalability of supply chain management.

2.2 Specific Objectives

- To develop an infrared-based line-following system that will allow the robot to follow predetermined paths in the warehouse layout.
- To incorporate RFID tag data into Firebase to support RFID-based junction detection for decision point computation and to permit task-specific activities (e.g., object placement or picking).
- To integrate an ultrasonic obstacle avoidance and detection technology that will allow the robot to automatically reroute and still be operational while safe, and detect obstacles up to 20 cm.
- To create and experiment with a recovery system that, when the line is lost, enables automatic recovery of the robot to the navigation path.
- To experiment with the reliability of the system as a whole, rate of successful recovery, reaction to obstacles, and positioning accuracy in warehouse-like environments.

2.3 Anticipated Outcomes of Objectives

1. Excellent Line-Following Ability

The robot shall have the ability to move continuously along straight, curved, and junction parts following black tape strips accurately.

The robot shall be able to accurately detect decision points and perform pre-decided operations with the aid of RFID tags for Reliable Junction Handling. It shall be able to synchronize with Firebase for task allocation as well.

2. Effective Obstacle Avoidance

Since the ultrasonic sensor system can detect obstructions within a pre-configured range (~20 cm), rerouting or braking can be done safely with a smooth process without compromising the system.

3. Effectively Recovering from Line Loss

In case of line loss, the recovery process will allow the robot to re-synchronize with the navigation route, saving on downtime and enhancing autonomy.

4. Navigation with the Cloud Integration

Real-time monitoring, assignment of tasks, and logging will be possible through real-time synchronization with Firebase to allow scalability to warehouse multi-robot systems.

5. System Performance Guaranteed

Comparisons of the system with traditional low-cost navigation methods, experimental demonstration will show quantifiable increments in navigation accuracy, recovery rate, obstruction handling efficiency, and reliability.

3. METHODOLOGY

3.1 System Diagram

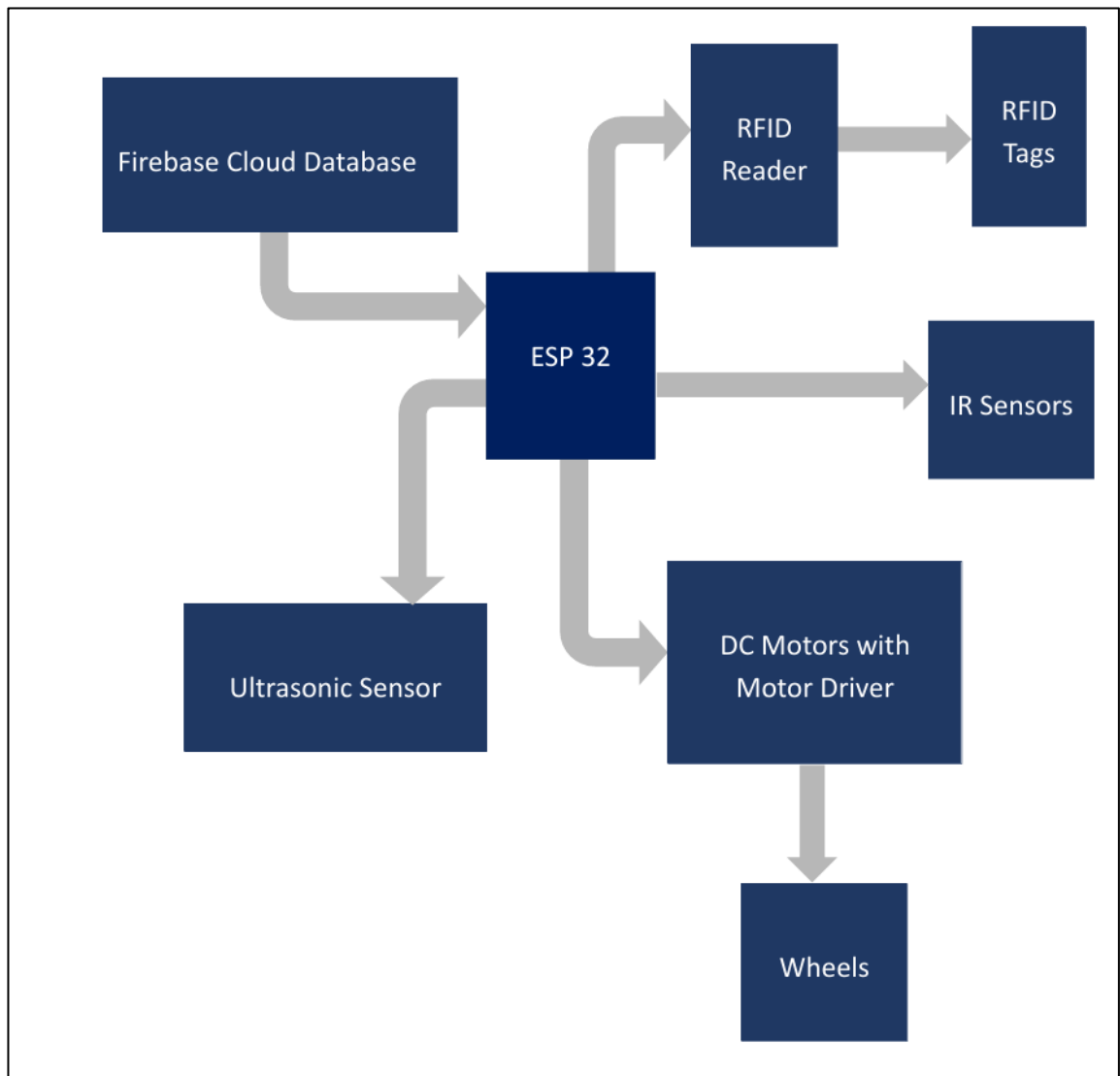


Figure 3. 1: System Overview Diagram

3.2 Implementation

Line-Following Module

IR sensors to be instructed to differentiate between white (floor) and black (line).

Logic:

- If center sensor is reading black → Go ahead.
- If left sensor is reading black → Turn right.
- If right sensor is reading black → Turn left.
- If all the sensors read white → Triggers the recovery mechanism.

RFID-Based Junction Detection

- The RFID tags are sensed by an RFID reader.
- Compared the read UID with the firebase data.
- The robot performs the appropriate action (i.e., halt for pick-up or drop-off) when the UID matches the location of a planned task.
- Obstacle detection using Ultrasonic Sensor
- The sensor calculates the distance to the obstacle.
- If object <20 cm: Stop → Reverse → Sidestep → Realign to path.

Recovery Mechanism

- The robot makes slow left-right turn during a line loss.
- Infrared sensors scan for the line.
- Robot realigns and proceeds after detection.

3.4 Tasks and Sub Tasks

3.4.1 Navigation Process

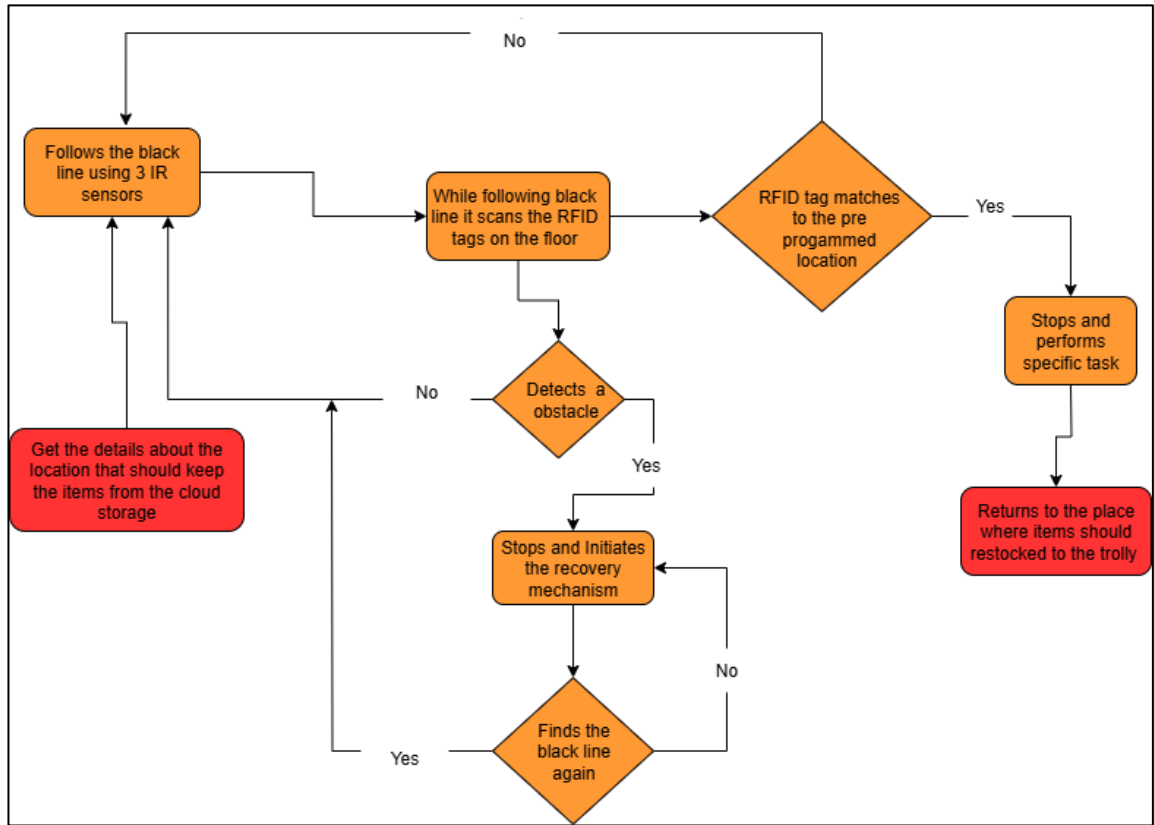


Figure 3.4. 1: Diagram of Navigation Process

Task 1: System Design

Subtask 1.1 : Identify the functional and non-functional requirements .

Subtask 1.2 : Hardware architecture design (microcontroller, sensors, chassis, and power system).

Subtask 1.3 : Design the software architecture (navigation logic, obstacle avoidance, recovery, and cloud integration).

Subtask 1.4 : Design flowcharts and system block diagrams.

Task 2: Hardware Development

Subtask 2.1: Buy components (ESP32/Arduino, motor driver, chassis, battery, RFID reader, ultrasonic sensor, and infrared sensors). Subtask 2.2: Construct a wheeled robotic chassis with motors.

Subtask 2.3: Mount and calibrate infrared sensors.

Subtask 2.4: Integrate RFID reader and tags.

Subtask 2.5: Installation and configuration of the ultrasound sensor.

Subtask 2.6: Testing power management (battery life, number of charge cycles).

Task 3: Software development

Subtask 3.1: Implement an IR line-following algorithm.

Subtask 3.2: Use Firebase mapping to support RFID-based junction detection.

Subtask 3.3: Run the obstacle detection and avoidance algorithm.

Subtask 3.4: Develop a line recovery system.

Subtask 3.5: Create a single, combined navigation software by combining everything.

Subtask 3.6: Implement Firebase connectivity (retrieving tasks, real-time sync).

Task 4: Integration

Subtask 4.1: Integrate the hardware and software.

Subtask 4.2: Perform basic calibration, debugging, and cabling.

Subtask 4.3: Study sensor fusion (harmonization of infrared, RFID, and ultrasonic).

Subtask 4.4: Make Firebase and cloud communication stable.

Task 5: Testing

Subtask 5.1: Unit test of the standalone module (IR, RFID, ultrasonic, Firebase).

Subtask 5.2: Test combined module integration.

Subtask 5.3: Stress test in a dynamic warehouse with variable barriers, multitasking, and spurious connectivity.

Task 6: Documentation

Subtask 6.1: Keep record of design and implementation stages in a project diary.

Subtask 6.2: Summarize lengthy test results and methodology for the study report.

Subtask 6.3: Summarize and state any limitations.

Subtask 6.4: Complete the final project report, conclusions, and recommendations.

Task 7: Assessment

Subtask 7.1: Estimate performance measures (accuracy, response time, and success rate).

Subtask 7.2: Contrast results with objectives.

Subtask 7.3: Report results and suggest improvements.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

The following functional goals are supposed to be accomplished by the CargoMate Smart Robotic Inventory System:

1. Using Infrared Sensors for Line Following

The robot has to follow predefined, black tape-marked warehouse routes.

Three infrared sensors (left, center, and right) should be used to constantly adjust movement and maintain alignment with the path.

2. Junction Detection Using RFID

The robot must recognize embedded RFID tags at decision points. The required action (select, place, or continue) should be determined by matching the unique ID of each scanned tag with Firebase data.

3. Detect Obstacles

The CargoMate must use its ultrasonic sensor to detect objects at 20 cm or nearer.

If it sees an obstacle, then it should stop, side-step (back up), and go back to the black line.

4. Recovering Over Line-Loss

The robot ought to stop and execute a search-and-realign procedure if it gets off track. The robot needs the IR sensors to assist it find the black line again and start navigating on its own.

4.2 Non Functional Requirements

For usability and stability assurance, the CargoMate Smart Robotic Inventory System must also meet the following non-functional requirements:

1. Availability

The gadget should be capable of working effectively in standard warehouse environment for at least two hours.

2. Scalability

Several CargoMate robots in the same warehouse should be capable of integrating due to the architecture, sharing tasks, and data using Firebase.

3. Cost-efficiency

To ensure the solution is economical for small and medium enterprises as well as scalable warehouses, the design should rely on cheap sensors and hardware (infrared, RFID, ultrasonic, microcontrollers).

4. Maintainability

The system design can make use of modular hardware and structured code to facilitate simple replacement, upgrade, and debugging of components.

5. Performance

To enable smooth and effective operation, the robot should reply to sensor information and make navigation decisions in real time (with little delay).

4.3 User Requirements

1. Usability:

The robot should be easy to use and not require extensive training of warehouse personnel.

2. Independence:

Without human intervention, on its own, the robot should be able to navigate, detect crossroads, and navigate around obstructions.

3. Recovery from Errors:

Instead of requiring manual reset when it gets lost on its route, the robot should be capable of recovering itself in case it gets lost.

4. Reliability:

When the component or sensor fails, its replacement must be easily accessible to warehouse personnel.

4.4 System Requirements

1. Hardware Requirements

- ESP32
- IR Sensors(3)
- RFID Reader + RFID Tags
- Ultrasonic Sensor
- Black Tape
- DC Motors with Motor Driver

2. Software Requirements

- Arduino IDE

4.5 Test Cases

Test Case 1 : Following a Straight Line

Goal: Use infrared sensors to make CargoMate always follow a straight black line.

Preparation: Position the robot at the start of a straight, uninterrupted black tape line.

Input: Switch the robot's power on and set it to navigation mode.

Desired outcome: The robot should use its infrared sensors to sense the middle line and proceed straight ahead without veering.

Test Case 2 : Identifying and Avoiding Obstacles

Objective: Verify whether the ultrasonic sensor identifies obstacles and initiates avoidance properly.

Configuration: Set a box or similar hard object 15 cm ahead of the robot on the black line.

Input: Move towards the obstacle.

Expected Result: CargoMate should stop, back up a bit, avoid the obstacle, and re-orient along the black line to proceed after crossing the 20 cm threshold.

Test Case 3 : Recovery after Line-Loss

Objective: Test the recovery system of the robot if lost sight of the line.

Configuration: Position the robot where the black line is deliberately missing or broken.

Input: Let the robot proceed with its navigation until it can no longer perceive the line.

Desired Outcome: In continuing navigation, CargoMate should stop, progressively turn left to right using infrared sensors in searching for the line, then align.

4.6 Use Cases

Use Case 1: Line Following Transportation of Goods

Actor: Autonomous robot

Precondition: The robot is positioned with a pre-programmed black line pattern on the warehouse ground.

Situation:

1. The robot detects the line with infrared sensors.

2. It travels along the route to arrive at a target point (such as a dispatch point or storage rack).
3. It halts for pickup and drop-off as it arrives.

Result: The robot consistently arrives at the destination without straying.

Use Case 2: Junction Identification and Decision-Making

Actor: Firebase (cloud infrastructure) + autonomous robot

Prerequisite: RFID tags are installed at intersections on the warehouse floor.

Situation:

1. An intersection with multiple potential paths is navigated by the robot.
2. The tag is read by an RFID reader.
3. To find the next goal or location, Firebase is asked about by the robot.
4. It moves in the right direction on the basis of information.

Result: The path of the robot dynamically changes according to available inventory requirements.

Use Case 3: Avoidance and Detection of Obstacles

Actor: Autonomous robot

Precondition: A worker or package is seen within 20 cm of the robot as it navigates in the warehouse.

Situation:

1. The obstacle is detected with an ultrasonic sensor.
2. The robot stops immediately.
3. Performs the avoidance drill (reverse → sidestep → rejoin line).
4. Reverts back to navigating the line.

Result: The robot resumes where it left off without colliding with objects.

Use Case 4: Line Recovery Mechanism

Actor: Autonomous robot

Precondition: The robot inadvertently loses the line (because of a disturbance or reflection on the floor).

Situation:

1. The robot halts and goes into recovery mode.
2. As IR sensors find the line, it keeps rotating left and right step by step.
3. It aligns and moves further after it detects the line.

Result: The robot navigates on its own without any intervention by humans.

Use Case 5: Task Delegation and Cloud Monitoring

Actor: Robot + Firebase (Cloud Platform)

Prerequisite: Robot's Wi-Fi connectivity to Firebase.

Situation :

1. Firebase receives real-time navigation data (status, location) from the robot.
2. A new task (e.g., proceed to RFID tag "SHELF-12") is assigned by the warehouse administrator.
3. Firebase provides instructions.
4. The robot executes using navigation logic.

Result: Real-time task assignment and tracking provide scalable operation.

5.PROJECT MILESTONES

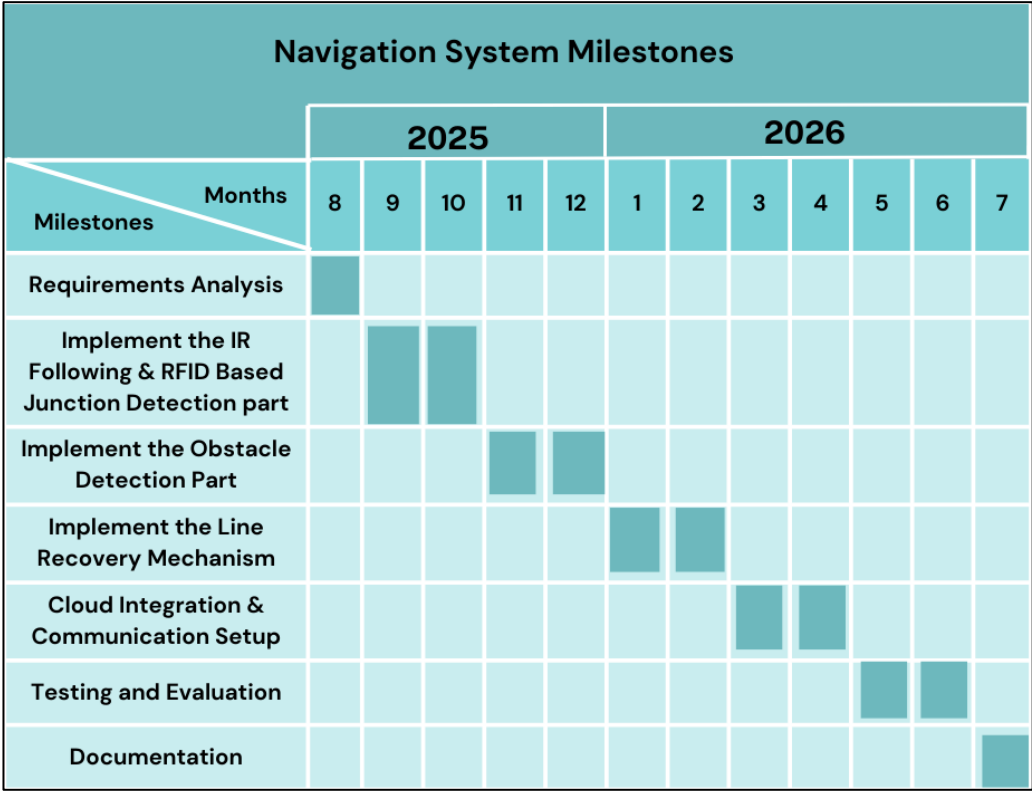


Figure 5. 1 : Gannt Chart

6. DESCRIPTION OF PERSONAL AND FACILITIES

Facilitators

Mr. Uditha Dharmakeerthi Sri Lanka Institute of Information Technology(SLIIT)	Prof. Anuradha Jayakody Sri Lanka Institute of Information Technology(SLIIT)
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7. BUDGET AND BUDGET JUSTIFICATION

Table 7. 1: Table of the budget

Item	Cost(Rs.)	Justification
IR Sensors (3)	300	Required for line-following operation. Three sensors enhance path detection stability and accuracy.
RFID Reader + RFID Tags	2,500	Permits junction detection and location identification, enabling the robot to move across the warehouse with intelligence.
Ultrasonic Sensor	300	Ensures safety and seamless operation in dynamic conditions through obstacle detection and avoidance.
ESP32	1,100	Acts as the primary controller via Bluetooth and Wi-Fi, facilitating computing and IoT/cloud connectivity.
Motor Driver	650	Interfaces allowing for stable direction and speed control from motors to ESP32.
DC Motors	1,500	Give the robot enough mobility and torque to move steadily and handle light loads.
Wheels	300	Make sure you have enough grip and stability to ride smoothly over the warehouse floor.
Wires	500	Needed to minimize the risk of failure by providing reliable electrical connections between parts.
Total	7150	

8.COMMERCIALIZATION

CargoMate is a cost-effective, scalable warehouse automation solution for big e-commerce order fulfilment warehouses, schools, and small and medium-sized businesses (SMEs). Strong autonomous navigation is enabled by its hybrid of ultrasonic obstacle detection, RFID junction sensing, infrared line following and recovery mechanism. Remote supervision, multi-robot cooperation, and task tracking are enabled by Firebase's real-time cloud integration. CargoMate has a good chance of being adopted in warehouses looking for cost-effective, safe, and successful automation because of its modular hardware, simple deployment, and low cost of manufacture.

8.1 Target Audience

1. Small and Medium-Sized Enterprises (SMEs): With its price quite astronomical, the majority of SMEs cannot incorporate state-of-the-art AGVs (Automated Guided Vehicles) or warehouse robots. CargoMate is an affordable, scalable alternative.
2. Warehouses and e-commerce order fulfilment facilities can use CargoMate in quantities for recurrent pick-and-place procedures, minimizing human labour needs.
3. Universities and research centres: Affordable solution for training in automation and research labs.

8.2 Competitive Advantage

1. Competes on the basis of providing a low-cost "light automation" solution versus expensive AGVs.
2. Modular design: Hardware and software can be upgraded (e.g., controlling multiple robots through the cloud).
3. Redundant hybrid navigation comes at a low cost through the combination of RFID decision points, ultrasonic obstacle control, and infrared path following.

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